

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)	I	{is complimentary to/attaches to} the <u>start</u> of the sequence/ allows binding of {taq/DNA} polymerase	1			1		
			II	breaks hydrogen bonds between strands/ separating the two strands/ DNA becomes single stranded	1			1		
			III	to allow the primers to { bind / anneal} (to the DNA strands)	1			1		
		(ii)		thermally stable/can withstand {high temperatures/70°C+}/ resistant to heat/eq	1			1		
	(b)			Any two (×1) from: - Populations are geographically isolated/ or description of (1) - Different environments select different alleles/ genetic drift / no gene flow (1) (In each area) they can interbreed and so {remain genetically close/ gene pools remain similar}/ ORA (1)		2		2		
	(c)			Elephants with tusks would be (more likely to be) {poached/killed}/ greater <u>selection pressure</u> on tusked elephants/ ORA (1) {Poaching/killing} is selecting {for tuskless elephants/against tusked}/ tuskless elephants have a <u>selective</u> advantage/ ORA (1) so {pass on their {alleles/ mutation} (for no tusks)/ increased frequency of tuskless alleles in gene pool / ORA (1)		1	2	3		